Homework 5

Math 2031 — Professor Hamilton

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Section 5.4

Question 2

Statement: Let $n \in \mathbb{Z}, n > 1$. Prove that if n is not divisible by any prime number less than or equal to $\sqrt{n}$, then n is a prime number.

Proof by Contradiction:

Suppose not.
Suppose an integer $n > 1$ is not divisible by any prime number less than or equal to $\sqrt{n}$ and n is not a prime number, ergo composite.
By definition, since n is composite, $n = yk_1$ for some intgers y, k_1.
Moreover: $1 < y \leq k_1 < n$.
By assumption, since n is not divisible by any prime less than or equal to $\sqrt{n}$ then:
$y > \sqrt{n} \wedge k > \sqrt{n}$:
However, this would lead to:
$y \cdot k_1 > \sqrt{n} \cdot \sqrt{n} > n$
Which is a contradiction since we assumed $y \cdot k_1 = n$ and therefore the smaller factor, y must be less than or equal to $\sqrt{n}$.
$\therefore$ since the negation leads to contradiction, the original statement must be true.
If n is not divisible by any prime number less than or equal to $\sqrt{n}$, then n is a prime number.
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Question 6a

Statement: Prove that $\sqrt{3}$ is irrational.

Proof by Contradiction:

Suppose not. Suppose that $\sqrt{3}$ is rational.
Therefore by definition, $\sqrt{3} = \frac{a}{b}$ for some integers a, b where $b > 0$.
Assume also that $\frac{a}{b}$ are in their simplified form, meaning $\gcd(a, b) = 1$.

Squaring both sides we get:
$3 = \left(\frac{a}{b}\right)^2 = \frac{a^2}{b^2}$
This allows us to express a^2 as:
$a^2 = 3b^2$
Thus we get that $3|a^2 \rightarrow 3|a \cdot a$ and by Euclid's lemma $\rightarrow 3|a$.
Therefore we get:
$a = 3k$ for some integer k. Going back and substituting to our expression above: $(3k)^2 = 3b^2$
Simplifying:
$9k^2 = 3b^2$
Thus:
$b^2 = 3k^2$
And again as per the previous steps, we get $3|b$.
Therefore since, $3|a$ and $3|b$ we get that $\gcd(a, b) \geq 3$.
However, by our original assumption, $\gcd(a, b) = 1$ and $1 \geq 3$ is a contradiction.
Therefore, since the negation leads to a contradiction, it must follow that the original statement is true.
$\sqrt{3}$ is irrational.
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Question 9

Statement: Let $n \in \mathbb{Z}, n \geq 1$. Prove that n is a perfect square if and only if, when n is written in standard form, all of the exponents are even.

Proof:

Part 1: n is a perfect square $\rightarrow$ n's standard form exponents are all even.

Let $n \in \mathbb{Z}, n \geq 1$ such that n is a perfect square thus $n = k^2$ for some integer k.
We want to show when n is written in standard form, all of the exponents are even.
Writing m in standard form we get:
$k = p_1^{m_1} \cdot p_2^{m_2} \cdot ... \cdot p_r^{m_r}$
We know $n = k^2$ therefore:
$n = k^2 = (p_1^{m_1} \cdot p_2^{m_2} \cdot ... \cdot p_r^{m_r})^2$
Applying the power rule for exponents:
$n = k^2 = p_1^{2m_1} \cdot p_2^{2m_2} \cdot ... \cdot p_r^{2m_r}$
Therefore, since $2|2m_i$ for all $1 \leq i \leq r$ thus by definition all the exponents are even.

Part 2: Standard form exponents of n are all even $\rightarrow$ n is a perfect square.

Let $n \in \mathbb{Z}, n \geq 1$ such that all the standard form exponents of n are even.
We want to show thus that n is a perfect square.
Writing out n in standard form including our assumption:
$n = p_1^{2m_1} \cdot p_2^{2m_2} \cdot ... \cdot p_r^{2m_r}$

Let $k^2 = n$.
Thus:
$k^2 = p_1^{2m_1} \cdot p_2^{2m_2} \cdot ... \cdot p_r^{2m_r}$
Taking the square root of both sides:
$k = \left(p_1^{2m_1} \cdot p_2^{2m_2} \cdot ... \cdot p_r^{2m_r}\right)^{\frac{1}{2}}$
Applying the power of exponents once more:
$k = p_1^{m_1} \cdot p_2^{m_2} \cdot ... \cdot p_r^{m_r}$
Note here that k is still an integer, since integers are closed through multiplication, moreover, $\forall p_i, m_i \in \mathbb{Z}$.
Therefore, since k is an integer, and $n = k^2$, thus by definition it is a perfect square as it is the square of another integer.
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Thus we have demonstrated that $\forall n \in \mathbb{Z}^+, n$ is a perfect square $\leftrightarrow$ all the exponents of n are even in its standard form.
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Question 13

Statement: Prove that 2 is the only prime of the form $n^3 + 1$.

Proof:

Let $n \in \mathbb{Z}$.
We want to show that $n^3 + 1$ is prime only for 2.
Using the sum of cubes formula we get:
$n^3 + 1 = (n + 1)(n^2 - n + 1)$
By the definition of any prime integer, at least one of its divisors must be 1 or -1.

Case 1: $n + 1 = 1$

This case forces $n + 1 = 1 \rightarrow n = 0$ which gives us the equation:
$0^3 + 1 = 1$
And by definition, 1 is not a prime number.

Case 2: $n + 1 = -1$

This case forces $n + 1 = -1 \rightarrow n = -2$ which gives us the equation:
$(-2)^3 + 1 = -8 + 1$
Again, -7 is not a prime number.

Case 3: $n^2 - n + 1 = 1$

This case simplifies to: $n^2 - n = 0 \rightarrow n(n - 1) = 0 \rightarrow n = 0 \vee n = 1$
For $n = 0$ see *Case 1.*
For $n = 1$, we get $1^3 + 1 = 2$ which is a prime integer.

Case 4: $n^2 - n + 1 = -1$

This case simplifies to $n^2 - n + 2 = 0 \to D = (-1)^2 - 4(1)(2) = -7$.
Since the discriminant of the equation is -7, there are no real solutions to our quadratic equation.
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Since 2 was the only possible prime integer that satisfied our equation, then by definition 2 is the only prime of the form $n^3 + 1$.
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Additional Problems

Question 1

Statement: Let a and b be natural numbers. Prove that if $a + b$ is prime, then $\gcd(a, b) = 1$.

Proof by Contradiction:

Suppose not. Let $a, b \in \mathbb{N}$ such that $a + b$ is prime and $\gcd(a, b) > 1$.
Let $d = \gcd(a, b)$.
Thus by definition, $a = dk_1$ and $b = dk_2$.
Thus we get that:
$a + b = dk_1 + dk_2 = d(k_1 + k_2)$.
Note here that we originally assumed $a + b$ is a prime number, thus the set of its divisors, S, is: $\{1, -1, (a + b), -(a + b)\}$.
However, we have demonstrated that $d|(a + b)$, moreover $d \notin S$ since $d > 1$ and $d \neq (a + b)$ as $d = \frac{a+b}{k_1+k_2}$.
Thus we have encountered a contradiction, since we assumed $a + b$ was prime number however $d|(a + b)$.
Therefore, since the negation leads to contradiction, the original statement must be true.
If $a + b$ is prime, then $\gcd(a, b) = 1$.
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Question 2

Statement: If $p \neq 5$ is an odd prime, prove that either $p^2 - 1$ or $p^2 + 1$ is divisible by 10.

Proof:

Let $p \neq 5$ be an odd prime.
We want to prove that either $10|p^2 - 1$ or $10|p^2 + 1$.
By the division algorithm, we get that:
$p = 10q + r$ for some integers q, r where $r \in \{1, 3, 7, 9\}$.
Note: We know $r \in \{1, 3, 7, 9\}$ since $0 \leq r < 10$, p is odd, moreover, $p \neq 5$, therefore it would not be within the remainder.

Thus, by congruence we get:
$p \equiv 1, 3, 7, 9 \bmod(10)$
Squaring both sides our congruence would form into:
$p^2 \equiv 1, 9 \bmod(10)$
For the next cases, instead of 9, we will make use of -1 by shifting as:
$-1 \equiv 9 \bmod(10)$.

Case 1: $p^2 - 1$

Given $p^2 \equiv 1, -1 \bmod(10)$, by definition $p^2 - 1 = 10k_1$ and $p^2 + 1 = 10k_2$ for some integers k_1, k_2.
For this case, we have demonstrated the conclusion as $p^2 - 1 = 10k_1$.

Case 2: $p^2 + 1$

Same, as case above, instead for $+1$ instead of -1.
$\therefore$ If $p \neq 5$ is an odd prime, prove that either $p^2 - 1$ or $p^2 + 1$ is divisible by 10.
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